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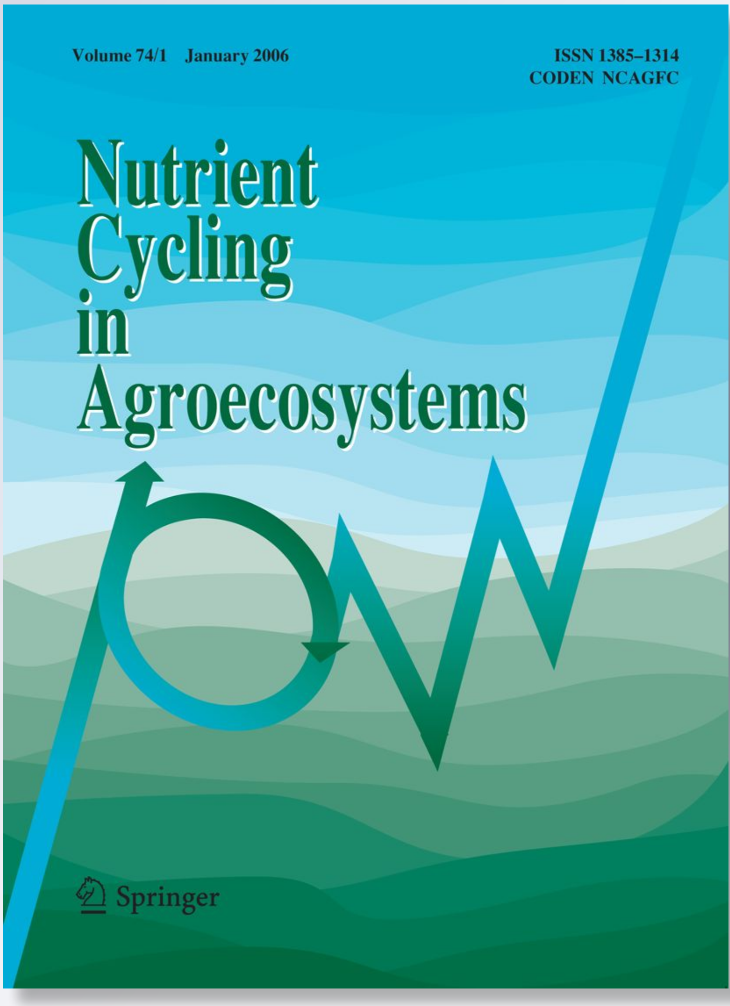

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ORIGINAL ARTICLE

A urine patch framework to simulate nitrogen leaching on New Zealand dairy farms

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Abstract On New Zealand dairy farms, it is the nitrogen excreted directly onto pasture, particularly urine, that drives nitrogen (N) leaching from the farm. A new framework (UPF: Urine Patch Framework) is presented that post-processes the results of a whole farm model and runs a mechanistic soil model to simulate the urine patches. Two alternative methods to simulate the spatial distribution of urine patches were implemented and compared (*Grid*: spatially explicit, and *Probabilistic*: based on the probability of different temporal urination patterns). This paper describes the implementation of these two methods in connection with a Whole Farm Model; and compares the N leaching predictions with observed data. Two examples are provided, one analyzing the impact of urine patch overlap and another, the relative risk of N leaching at different times of urinary N deposition. The model showed good correlation and predictive ability between simulated annual N leaching results and observed data [$R^2 = 94$ %, mean relative prediction error (MRPE) = 10 % for *Grid* and $R^2 = 72$ %, MRPE = 20 % for *Probabilistic*]. The two methods produced similar results across an 8-year period for monthly and annual N leaching ($R^2 = 96$ %, MRPE = 10 % and $R^2 = 86$ %, MRPE = 8 %; respectively). Only 8 % of the paddock area was covered with multiple urinations during 1 year, but as much as 39 % of the total urine volume was deposited on overlapped patches. Systematically removing all urinary N for 1 month in either May or June reduced N leaching by approximately 20 %. Avoiding urinary N deposition during autumn or early winter could be highly effective in mitigating N leached during the following winter.

Keywords Whole farm model · Soil model · Nitrogen excretion · Nitrogen leaching

Introduction

Agricultural fertiliser, stock manure and urine are the major non-point-sources of nitrogen (N) in rural New Zealand (Ministry for the Environment 2007). On New Zealand pastoral dairy farms, it is the nitrogen excreted on pasture, particularly urine patches, that drives environmental impact, and not nitrogen fertilizer inputs per se (Decau et al. 2004). This has wide relevance beyond New Zealand dairy farms, since the majority of the world's ruminants still urinate at pasture rather than inside a building. Furthermore, managed grazing is the single most extensive form of land use on the planet (Asner et al. 2004). Even for extensive systems where rotational grazing is not practiced, there are still important issues of animal concentration around water holes and camping areas

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where animals congregate (Beukes and Ellis 2003; Betteridge et al. 2010).

For a given soil type and climate, the volume of urine excreted onto pastures, the N concentration in the urine and the timing of the urine deposition are the main determinants of N losses from pastoral dairy farms. These elements vary strongly from farm to farm depending on management decisions about stocking rate, grazing management, supplementary feeding, calving dates, stock sales and purchases and the number of hours cows graze. The novel framework presented here consists of running an existing dairy whole farm model (WFM, Beukes et al. 2010b), and then automatically post-processing the outputs of the WFM to simulate urine patches using a mechanistic soil model (Probert et al. 1998) under the agricultural production systems simulator (APSIM, Keating et al. 2003). The WFM simulates individual cows and paddocks on daily time steps, while feed intake and diet composition (particularly N concentration and N fractions) change from day to day, according to user-entered decision rules controlling herbage allowance and supplementary feeding. The strength of this scheme is that it makes the most of the detailed information generated by the WFM, and therefore, can concurrently capture the variability among New Zealand dairy farm systems, and predict N leaching by using a detailed soil model.

Five elements complicate the modelling and prediction of N leaching at the farm level. First, excreta are not distributed evenly, but are deposited in patches with a high N concentration (Di and Cameron 2002). Secondly, the patches can overlap, which has been demonstrated to be important and has been considered in previous modelling studies (de Klein 2001; Vellinga et al. 2001, McGechan and Topp 2004; Pleasants et al. 2006). Shorten and Pleasants (2007) estimated that a double urine patch can leach three times more N than a single patch. Thirdly, the fate of the nitrogen from excreta strongly depends on the time it is deposited (Haynes and Williams 1993; Snow et al. 2007), in relation to climate conditions at deposition and the following months. Fourth, the effect of a single urination on soil and pasture can last for several months after deposition. Finally, grazing all year round is the norm in New Zealand dairy farms, and each paddock is grazed many times each year.

One option to represent such complexity at paddock level is a spatially explicit method (*Grid* method hereafter). This involves dividing the paddock into cells representing some proportion of the size of a typical urine patch, and allocating the urinations randomly for each grazing event (i.e. each time the animals visit a paddock). A second alternative is to calculate the probability of the different temporal urination patterns, estimate the N leaching from each pattern and then calculate the leaching at paddock level (*Probabilistic* method hereafter). Different simplifications are required for each method, to make implementation feasible.

The two objectives of this study were to describe the new framework, including the simplifications and assumptions required to implement the two methods and to compare N leaching predictions with measured data for a Waikato (New Zealand) dairy farm. Two simulated examples are reported to illustrate the application of the framework, by analysing the effect of overlapping urinations and the relative risk of N leaching at different times of urinary N deposition.

Materials and methods

The WFM, described by Beukes et al. (2008) represents a pasture-based dairy farm. Individual paddocks (using a simple pasture and soil water model, Romera et al. 2009) and individual cows (Molly: Baldwin 1995; Hanigan et al. 2009) are represented with daily time steps. This allows the user to represent a wide variety of management strategies, by combining decision rules to determine policies for: rotational grazing, supplementary feeding, fertilizer applications, irrigation (water and dairy effluent), cropping, surplus herbage conservation, reproduction, standing off cows, culling, replacement and mob (sub-group within the whole herd on the farm) assignment. The WFM was developed to assist in analysing and designing farm systems experiments, and to explore various scenarios requiring numerous farm system interactions over multiple years (Beukes et al. 2010a). The partitioning of excreted N is based on the proportion of the daily active time cows spend on each surface (pasture, standoff pad, dairy, races and concrete pad).

The modelling approach used a new framework (UPF: Urine Patch Framework) that post-processes the results of the WFM and runs a detailed soil model (APSIM: Keating et al. 2003) to simulate the urine patches (see section Urine patch in APSIM). The process involves the following steps:

1. The user runs the WFM for at least 2 years, collecting urine deposition information, plus other management events at a paddock level.
2. The WFM extracts the urine information for each grazing event on each paddock [Number of urinations, amount of N excreted (kg N), urine volume (Litres) and paddock area (ha)]. The WFM also creates event schedules for all the walkways, the milking parlour, and the stand-off pad to account for the partitioning of urine and faeces between these areas. Only the fate of urinary N directly excreted onto pastures is analyzed and reported here.
3. The UPF divides each paddock into cells: (*Grid* method) or areas with the same urination patterns (i.e. dates of deposition and amount excreted on each date, *Probabilistic* method).
4. The UPF generates an xml simulation file (.apsim) for each cell or pattern. The files contain the urine applications (time and N rate), defoliation (time and post grazing residual), N fertilizer applications (time and N rate), and the soil parameters.
5. The UPF invokes APSIM to run the simulation files.
6. The UPF collates the leaching data/results per paddock.

The WFM and the UPF run on normal PCs, and the user can select either method (*Grid* or *Probabilistic*) by changing one parameter in an initialisation file. The following sections describe the approach to link the WFM with APSIM.

General details

The following assumptions were applied across both methods:

- Uniform probability of urinations across the paddock, i.e. no camping (areas where the animals congregate to rest), no water trough effect, etc. (Moir et al. 2011).
- A paddock grazing event occurs over 1 day. In the WFM paddocks are frequently subdivided into strips (with temporary electric fences) which could be grazed over several consecutive days. When this is the case, the grazing event is allocated to the last day in the paddock.
- Faecal N input is ignored, which should have a minor effect on N leaching (Vellinga et al. 2001). This simplification can be relaxed in the future, which will be easier with the *Grid* method.
- The UPF post-processes WFM outputs and there is no feedback from the UPF to the WFM. Therefore, the differential pasture growth on the urine patches has no effect on animal response.
- The impact of urine deposition on selective grazing behaviour was ignored.

For a single cow, the daily urine volume (urine Volume, L cow^{-1} day^{-1}) was calculated using the following equation from Bates (2009), adapted from Bannink et al. (1999):

$$\begin{aligned}\mathrm{urineVolume} = 1.3441 + \mathrm{DMI} \times \; & (1.079\ \mathrm{Na}_{\%} \\ & + 0.538\ \mathrm{K}_{\%} + 0.1266\ \mathrm{N}_{\%}) \\ & - \mathrm{AMY} \times (0.126 + 0.0275\ \mathrm{MP}_{\%})\end{aligned} \tag{1}$$

where DMI (kg DM cow^{-1} day^{-1}): dry matter intake, $\mathrm{Na}_{\%}$: % of Na^{+} in the diet on a dry matter basis (default value 0.15 %); $\mathrm{K}_{\%}$: % of K^{+} in the diet on a dry matter basis (default value 2.5 %); $\mathrm{N}_{\%}$: % of N in the diet on a dry matter basis; AMY (kg cow^{-1} -day^{-1}): liquid milk yield; $\mathrm{MP}_{\%}$: % protein content of milk on a dry matter basis.

The total urine volume deposited during a grazing event (totalUrineVolume) is the sum of the urineVolume produced during that time by all the cows in the mob.

The total amount of urinary N produced in 1 day by the whole mob (totalUNexcretion, kg N day^{-1}) is calculated by the WFM from the variable NUrine in Molly (Johnson and Baldwin 2008).

Urine patch in APSIM

For each cell or pattern, APSIM reads the input data for soil characteristics, urine and nitrogen fertiliser applications, and defoliation events from xml files. Defoliation events are simulated as pasture cuts according to the residual herbage mass reported by the WFM.

The default modules in APSIM were used as described by Probert et al. (1998), namely SoilWat (soil water, www.apsim.info/wiki/SoilWat.ashx) and SoilN (carbon–nitrogen dynamics, www.apsim.info/wiki/SoilN.ashx) in this study. SoilWat is a cascading layer model, based on the CERES model, and simulates surface runoff, saturated and unsaturated water flow with the associated solute movement and soil evaporation. SoilN describes the dynamics of both carbon and nitrogen in soil. The organic matter is divided into two pools, one representing the more labile, soil microbial biomass and microbial products, and the other the rest of the organic matter (humus). The flows between pools are calculated in terms of carbon, the N flows depending on the C: N ratios. Mineralization or immobilization of mineral N (both ammonium-N and nitrate–N are available for immobilisation) is determined as the balance between the release of N during decomposition and immobilization during microbial synthesis and humification. Potential nitrification rate is assumed to follow Michaelis-Menton kinetics, and actual daily nitrification is reduced to allow for sub-optimal water, temperature and pH conditions. De-nitrification rate is estimated as a function of NO_3–N concentration, active carbon concentration, soil moisture and soil temperature.

Within APSIM, the pasture-crop was simulated using a version of the model described by Romera et al. (2009) and adapted for APSIM. Pasture growth depends on incident radiation, temperature and soil moisture, and in the adapted version on nitrogen content in the plant. Therefore, urinated and non-urinated areas would exhibit different herbage growth rates. The soil water and N uptake equations where adapted from the source code of the Root component of the Plant2 model (APSIM initiative, www.apsim.info/Wiki/).

The soil water supply to the pasture crop is simulated as:

$$\mathrm{SWSupply}_1 = \mathrm{kl}_1 \times (\mathrm{sw\_dep}_1 - \mathrm{ll}_{15}\mathrm{\_dep}_1) \tag{2}$$

where l: lth soil layer; kl_l: constant proportion of the available water that could be used in 1 day; sw_dep_l (mm): soil water content; $ll_{15}\_dep_l$ (mm): Lower limit soil water content (water retained to a soil potential of 15 bar).

The water uptake from each soil layer ($SWuptake_l$) is computed as:

$$\mathrm{SWuptake}_l = \frac{\sum_{l=1}^{n} \mathrm{SWSuppl}_l}{\mathrm{PET}} \times \mathrm{SWSuppl}_l \tag{3}$$

where PET (mm day^{-1}): potential evapotranspiration.

The N supply from the soil includes nitrate N (NO_3-N) and ammonium N (NH_4-N):

$$\mathrm{NO3Supply}_l = \text{minimum of } (\mathrm{no3}_l \times \mathrm{no3ppm}_l \times \mathrm{swaf}) \text{ and } \mathrm{no3}_l \tag{4}$$

where swaf: fraction of the potentially available water; $no3ppm_l$ (ppm): NO_3-N content; $no3_l$ (kg ha^{-1}): NO_3-N content.

$$\mathrm{NH4Supply}_l = \text{minimum of } (\mathrm{nh4}_l \times \mathrm{nh4ppm}_l \times \mathrm{swaf}) \text{ and } \mathrm{nh4}_l \tag{5}$$

where $NH4ppm_l$ (ppm): NH_4-N content. $NH4_l$ (kg ha^{-1}): NH_4-N content.

The N deficit in the plant, above and below ground, is as follows:

$$\mathrm{NDeficit} = (\mathrm{OptimumNConc}_A \times \mathrm{LiveWeight}_A - \mathrm{LiveN}_A) + (\mathrm{OptimumNConc}_R \times \mathrm{LiveWeight}_R - \mathrm{LiveN}_R) \tag{6}$$

where NDeficit (kg N ha^{-1}): nitrogen deficit in the pasture crop; A: above ground; R: roots; OptimumNConc (kg N kg DM^{-1}): optimum N concentration (0.04, 0.048, for above ground and roots respectively); LiveWeight (kg DM ha^{-1}): total N live material in the pasture crop; LiveN (kg N ha^{-1}): total N contained in the live material in the pasture crop.

The N uptake is calculated as:

$$\mathrm{NuptakeF} = \frac{\sum_{l=1}^{n} \mathrm{NO3Suppl}_l + \sum_{l=1}^{n} \mathrm{NH4Suppl}_l}{\mathrm{NDeficit}} \tag{7}$$

$$\mathrm{NO3uptake}_l = \mathrm{NuptakeF} \times \mathrm{NO3Supply}_l \tag{8}$$

$$\mathrm{NH4uptake}_l = \mathrm{NuptakeF} \times \mathrm{NH4Supply}_l \tag{9}$$

The effect of N deficit on herbage growth (McCallFn) is proportional to the N concentration in the plant relative to its optimum:

$$\mathrm{McCallFn} = \frac{(\mathrm{LiveNCon}_A - \mathrm{MinimumNConc}_A)}{\mathrm{OptimumNConc}_A - \mathrm{MinimumNConc}_A} \tag{10}$$

where $MinimumNConc_A$ (kg N kg DM^{-1}): minimum N concentration above ground (0.012); $LiveNConc_A$ (kg N ha^{-1}): N concentration in the live material above ground.

A constant proportion of the above ground growth as calculated by the pasture model (0.35) is allocated to the root growth. The total root growth is then divided among the root layers relatively to the depth of each layer and a relative root growth factor (XL_l) entered by the user. The roots lose a fraction of the DM through senescence each day (0.005), which is returned to the soil in the form of fresh organic matter (Probert et al. 1998).

In the following examples, the APSIM soil (Version 7.3) was parameterized to describe a Horotiu silt loam soil (Typic Orthic Allophanic soil; Singleton 1991) with seven layers (Table 1). Urine deposition was simulated in APSIM as nitrogen fertilizer plus 5 mm of water as irrigation. Nitrogen was applied in depth (as a first approximation, uniformly distributed among the first four soil layers) to represent preferential flow of urine through soil macropores. This is a simplification that will be reviewed in future versions, as the initial wetting pattern (area and depth) depends on the volume of the urination, soil type and soil moisture (Williams and Haynes 1994; Monaghan et al. 1999; Stout 2003). Nitrogen (in nitrate form) flow beyond layer 4 (55 cm) was considered as leaching, matching the depth at which observed data used to test the model were collected (Shepherd et al. 2010a). Perennial ryegrass, the dominant grass sown in New Zealand, is a shallow rooted species (Crush et al. 2005), for example Houlbrooke (1996) found approximately 80 % of the root mass in the top 7 cm of soil.

Once the APSIM soil model was parameterized to represent the Horotiu soil (Table 1), predicted N leaching and water drainage were compared against field data representing a single urine patch from Shepherd et al. (2010a). These authors used intact soil monolith lysimeters (300 mm diameter by 500 mm deep) encased in polyvinyl chloride (PVC) cylinders collected from a free draining Horotiu silt loam soil. Irrigation was applied at regular intervals to supplement rainfall, and to generate leaching equivalent to a typical winter in the Waikato region. Cow urine (equivalent to 500 kg N ha^{-1}) was applied in August 2005. Preliminary testing showed good agreement with lysimeter recorded N leaching and water drainage data (Fig. 1).

Grid method

For each paddock, the UPF creates a *Grid* with equal size cells. The area of each cell was set to one quarter of a urine patch. Each cell runs from the beginning to the end of the simulation. A necessary assumption for this method was a constant area per urination for all the grazing events throughout the simulation (0.5 m^2 in this study). The area of the cells and of the urine patches are input parameters that can be modified by the user.

The total number of cells (numberOfCells) per paddock was calculated as:

$$\text{numberOfCells} = \frac{\text{paddockArea} \times 10{,}000}{\text{areaPerUrination}} \times c \qquad (11)$$

where paddockArea (ha): area of the paddock; areaPerUrination: constant representing the area covered by one urination (default 0.5 m^2); 10,000: m^2 ha$^{-1;}$ c = cells per urine patch (default value 4), i.e. each cell has an area equivalent to 0.25 of the assumed urine patch area.

For example, if the size of the urine patch is 0.5 m^2 and the size of the paddock is 1 ha, then the number of cells required would be 80,000 [= 10,000/(0.5/4)].

The number of urinations per grazing event (totalNumOfUrinations) to be applied to the *Grid*, can be calculated as:

$$\text{totalNumofUrinations} = \frac{\text{totalUrinatedArea}}{\text{areaPerUrination}} \qquad (12)$$

where totalUrinatedArea (m^2): sum of the total area of all the patches in one grazing event.

The totalUrinatedArea is:

$$\text{totalUrinatedArea} = \frac{\text{totalUrineVolume}}{\text{urineColumn} \times 0.001}$$

totalUrineVolume (m^3): The total urine volume deposited on a grazing event, calculated in WFM; urineColumn (mm): constant representing the amount of water in the urine, equivalent to irrigation or rainfall (default value 5 mm, McGechan and Topp 2004); 0.001 = m mm^{-1}.

The N deposition rate on the urine patch (UNrateUG, kg N ha^{-1}) can be calculated as:

$$\text{UNrateUG} = \frac{\text{totalUNexcretion}}{\text{totalUrinatedArea}} \times 10{,}000 \qquad (13)$$

where totalUNexcretion (kg): total urinary N excretion during a grazing event; 10,000: m^2 ha^{-1}.

All the urination events are randomly applied to the cells of the *Grid*, generating different urination patterns. The number of cells to run can be pruned by grouping cells categorised by the same urination pattern and by collating all the non-urinated cells into one. The opportunities for pruning diminish as the number of grazing events increases.

Table 1 Inputs for the SOILWAT and SOILN model (Probert et al. 1998) to represent a Horotiu soil (V. Snow, personal communication)

Soil properties and initial values by layer								
Layer	1	2	3	4	5	6	7	Layer number
dlayer	60	110	140	240	180	180	160	Layer depth (mm)
ll_{15}	0.25	0.18	0.16	0.13	0.08	0.04	0.04	Lower limit soil water content (volumetric, mm mm^{-1})
air_dry	0.076	0.057	0.051	0.044	0.026	0.012	0.012	Water content to which soil can dry by evaporation (volumetric, mm mm^{-1})
dul	0.41	0.35	0.33	0.3	0.2	0.15	0.13	Drainage upper limit (volumetric, mm mm^{-1})
sat	0.593	0.606	0.535	0.5	0.497	0.444	0.369	Saturated water content (volumetric, mm mm^{-1})
sw	0.33	0.265	0.245	0.215	0.14	0.095	0.085	Initial soil water content (volumetric, mm mm^{-1})
bd	0.88	0.85	1.01	1.05	1.05	0.84	0.69	Bulk density (g cm^{-3})
oc	8	1.88	0.93	0.43	0.13	0.1	0.1	Organic carbon (%)
ph	6	6.43	6.72	6.8	6.37	6.41	6.43	Soil pH
nh4	33.334	22.5	1.389	1.389	1.389	1.389	1.389	Initial NH_4-N (ppm)
no3	0.1	0.1	0.1	0.1	0.1	0.1	0.1	Initial NO_3-N (ppm)
finert	0.6	0.7	0.8	0.9	0.99	1	1	Proportion of initial organic carbon assumed to be inert
fbiom	0.04	0.03	0.02	0.01	0.01	0.01	0.01	Initial labile soil biomass (biom) as proportion of non-inert carbon
swcon	0.8	0.8	0.8	0.8	0.8	0.8	0.8	Proportion of water above dul that drains each day

Soil water parameters		
u	9	Stage 1 soil evaporation coefficient (mm)
cona	4.4	Coefficient for stage 2 soil evaporation (mm $day^{-0.5}$)
cn2	68	Runoff curve number
cn_red	20	Maximum reduction in cn2 due to presence of surface residues
salb	0.18	Bare soil albedo
difuss_const	88	Coefficient defining diffusivity
difuss_slope	35.4	Coefficient defining diffusivity
soil_cn	11.5	C:N ratio of soil
root_wt	1500	Initial root mass (kg DM ha^{-1})
root_cnr	30	C:N in the root mass
residue_wt	800	Initial surface residue mass, i.e. pasture litter (kg DM ha^{-1})
residue_cnr	80	C:N ratio of surface residues

Probabilistic method

After a single grazing event, areas of the paddock can be categorised as; no urination (B), single urination (U) or multiple urinations (O) (i.e. 3 *categories*). After 2 grazing events, there are 9 possible combinations of categories (*patterns* hereafter), and after 3 grazing events, there will be 27 possible patterns ranging from BBB where an area receives no urine on each occasion, to OOO where an area receives urine more than once on each occasion.

On average, there could be around 12 grazing events per paddock in a year, which would create 3^{12} (531,441) patterns. Using the APSIM model, and considering a run time of 4 s per simulation year, it would take approximately 24 days to simulate one paddock for 1 year, or 492 days for a farm with 20 paddocks.

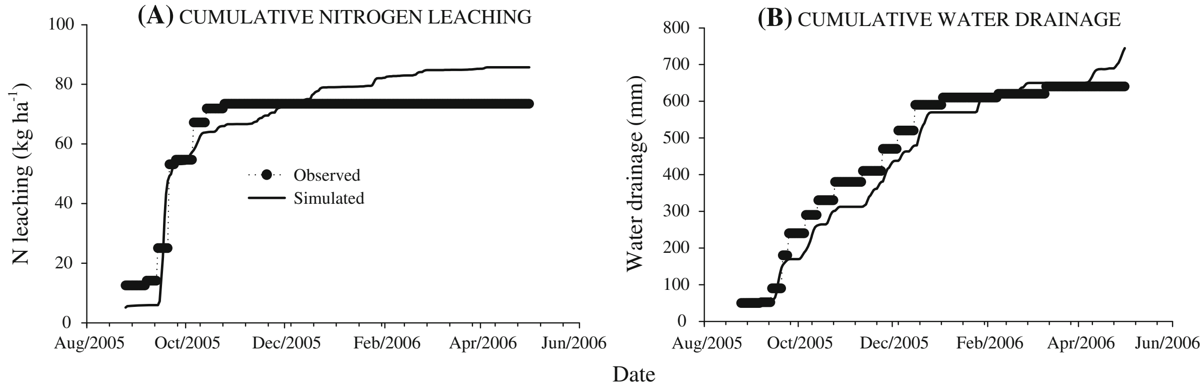

Fig. 1 Comparison between observed and simulated N **a** leaching and **b** cumulative water drainage from application of an artificial urine patch equivalent to 500 kg N ha^{-1} in August 2005

To reduce processing time, the *Probabilistic* method was simplified using the assumption that the effect of one urination event can be neglected a certain number of months after deposition (*months to remember* hereafter). This implied that the full simulation period (i.e. the WFM run) can be subdivided into periods of time (months to remember +1 month), which kept the exponential explosion in the number of urination patterns under control. A range from 6 to 12 months to remember was explored. For one period at a time, all the patterns are created, and later simulated in APSIM. Nitrogen leaching results are collected only in the last month of the period, whereas the previous months are used to prime the soil model with relevant historical information (climate, fertilization, defoliation and urination events).

The partitioning of simulation time in 7-month periods (6 months to remember), for a paddock with 12 grazing events over a 1-year period is given in Table 2 as an example. Considering 3 categories (B, U, and O) per grazing event, after 7 months (or grazing events in the example) there will be 2,187 different patterns. To cover 12 grazing events (i.e. 1 year), the UPF would simulate 26,244 runs (2,187 × 12), which would take approximately 0.7 days per paddock (reduced from 24 days as described above). The basic assumption in this example is that to estimate the N leaching on any given month, only the previous 6 months are relevant. Each one of the 2,187 patterns is run independently in APSIM for the period covered by 7 months (see Appendix for more details).

Other assumptions for the probabilistic method were:

- An average size of all urine patches within each grazing event was selected. Although, between grazing events, patch sizes do vary with urine volume per urination (approximately 0.2 m^2 per litre), which in turn is a function of cow size (see Eq. 15).
- Cows are simulated individually in the WFM and produce different amounts of urinary N, but in the UPF the N content of all urine patches within the same grazing event were considered equal (AveVolumePerUrination, see below).
- Overlapped patches (areas urinated upon more than once) within each grazing event are combined into one category (Eq. 20).
- The Poisson distribution (see de Klein 2001) was used to calculate the area associated with each urination pattern category.
- Patterns are sorted by probability and those that together account for less than 1 % of the urinated area are omitted. The urinary N in the omitted patterns is evenly distributed among the others, proportionally to their area.

Area per urination

The volumePerUrination was provisionally calculated by assuming that the frequently quoted volume of 2.5 L corresponds to an average New Zealand crossbred adult cow with a live weight (LW) of 486 kg. The

Table 2 Number of patterns to be simulated when working in 7 month periods (6 months to remember)

Periods	Pre-run						Month (n)											
	−6	−5	−4	−3	−2	−1	1	2	3	4	5	6	7	8	9	10	11	12
1	3	9	27	81	243	729	*2,187*											
2		3	9	27	81	243	729	*2,187*										
3			3	9	27	81	243	729	*2,187*									
4				3	9	27	81	243	729	*2,187*								
5					3	9	27	81	243	729	*2,187*							
6						3	9	27	81	243	729	*2,187*						
7							3	9	27	81	243	729	*2,187*					
8								3	9	27	81	243	729	*2,187*				
9									3	9	27	81	243	729	*2,187*			
10										3	9	27	81	243	729	*2,187*		
11											3	9	27	81	243	729	*2,187*	
12												3	9	27	81	243	729	*2,187*

Example of a paddock with 12 grazing events, one per month (i.e. covering 1 year). Assuming 3 categories for each grazing event: baseline (B, no urine), single urination (U) and multiple urinations (O). Only the N leaching results from the *italicized* cells are collected. All 2187 possible patterns on each period are simulated from month t − 6 up to month t single-urination-volume for any other adult cow is calculated as adultVolumePerUrination = 2.5/486 × MW (MW is mature weight, kg). The volumePer Urination (L) for young cows was assumed to be proportional to her metabolic weight ($LW^{0.75}$) relative to mature metabolic weight ($MW^{0.75}$):

$$\text{volumePerUrination} = \text{adultVolumePerUrination} \times \frac{LW^{0.75}}{MW^{0.75}} \quad (14)$$

The average volume per urination (AveVolumePerUrination) is the average for all the cows in the mob on a particular day.

The area covered by one urination (areaPerUrination, m^2 $urination^{-1}$) was calculated assuming a urineColumn of 5 mm and a cylinder shape. Knowing the volume per urination, it is possible to calculate the area affected by a single urination as:

$$\text{areaPerUrination} = \frac{\text{AveVolumePerUrination} \times \frac{1{,}000}{10{,}000}}{\text{urineColumn} \times 0.1} = \frac{\text{AveVolumePerUrination}}{\text{urineColumn}} \quad (15)$$

where: 1,000: cm^3 L^{-1}; 10,000: cm^2 m^{-2} and 0.1: cm mm^{-1}.

Paddock scale urination overlap and patterns

For each cow, each day, the total number of urinations is the result of:

$$\text{numUrinations} = \frac{\text{urineVolume}}{\text{volumePerUrination}} \quad (16)$$

where urineVolume (L): volume of urine produced during 1 day by a cow (see Eq. 1).

The total number of urinations (totalNumOfUrinations) is the sum of numUrination for all the cows in the mob.

The Poisson distribution was used to calculate the proportion of the area of the paddock covered by each category within a grazing event:

$$p(D, r) = \frac{e^{-D}(D)^r}{r!} \quad (17)$$

where D: urine density, which is the sum of the area of all the patches in one grazing event as a proportion of the total paddock area.

$$D = \frac{\text{totalNumOfUrinations} \times \text{areaPerUrination}}{\text{paddockArea}}$$

paddockArea (ha) = area of the paddock. r: number of excreta on the same area (0, 1, 2 … n), i.e. levels of overlapping. totalNumOfUrinations: total number of urinations during a grazing event (sum of the individual number of urination, see Eq. 16).

The proportion of the area with no recorded urination is $p_{(D,r=0)}$, for a single urinations the area is $p_{(D,r=1)}$ and for multiple urinations [i.e. $p_{(D,r>1)}$], the proportion was calculated as the remaining area: $p_{(D,r>1)} = 1 - p_{(D,r=0)} - p_{(D,r=1)}$.

When several grazing events are combined, the probability of each pattern was calculated as:

$$P(D_{1\ldots n}, r_{1\ldots n}) = \prod_{i=1}^{n} p(D_i, r_i) \quad (18)$$

where i: ith grazing event; n: number of grazing events being considered; r: number of urinations on the same area in the ith grazing event.

Urinary N deposition rate

Within one grazing event, the rate of urinary N applied on a single urination (UNrateUP; kg N ha^{-1} equivalent) depends on the average amount of N per urination and the area covered by a single urination.

$$\text{UNrateUP} = \frac{\text{totalUNexcretion}}{\text{totalNumOfUrinations} \times \text{areaPerUrination} \times 10{,}000} \quad (19)$$

where areaPerUrination (m^2): see Eq. 15. 10,000: m^2 ha^{-1}.

Since the area urinated upon more than once is to be collated, the N rate on such area (UNrateOP, kg N ha^{-1}) is calculated as:

$$\text{UNrateOP} = \frac{\text{totalUNexcretion} - \text{UNrateUP} \times \text{paddockArea} \times p_{(r=1)}}{\text{paddockArea} \times p_{(r>1)}} \quad (20)$$

The second term in the numerator [UNrateUP × paddockArea × $p_{(r=1)}$] is the amount of urinary N (kg) deposited as single urinations.

Model testing

A typical 'all pasture' Waikato dairy farm was set up in the WFM, with a stocking rate of 3.38 cow ha^{-1} (initial average live weight 483 ± 71 kg cow^{-1}), grazing all year round on perennial ryegrass based pastures. Nitrogen fertiliser was applied at a rate of 230 kg N ha^{-1} year^{-1}, spread over five applications. Paddocks were rotationally grazed (strip grazing with temporary electric fences), with grazing rounds of 30–100 days in winter; 20 days in spring; 30 days in summer and 40–80 days in autumn. The calving season was July–August and the herd was milked twice a day until the 10th of May when cows were dried off. Replacement cows were purchased as 2 year olds at the end of May. The Waikato region has a temperate climate, with an average temperature of 13.9°C and an annual rainfall of 1,105 mm during the study period (2001–2008).

The effect of the number of months to remember (from 6 to 12) in the *Probabilistic* method was analyzed for the year 2005, in terms of N leaching predictions and computing time on one sample paddock. The WMF and the UPF were run for the years 2004 and 2005, but only the second year was analyzed, i.e. after a meaningful history of urination patterns was established. To confirm the similarity of the N leaching results from both methods, the WFM and UPF were run for 8 pairs of years (2000–2001 to 2007–2008), always taking the second year for the comparisons.

The N leaching from the simulated farm described above was compared with the N leaching measured in Control farmlet (7 ha) described by Ledgard et al. (2006). No attempts were made to represent the real farmlet faithfully, therefore, only annual leaching patterns are of interest at this stage. Nitrogen leaching data were collected using porous ceramic cup collectors as described by Ledgard et al. (2006) (Table 3). Accumulated annual N leaching data from 2002 to 2008 were used to assess model predictions. The model results (N leaching below soil layer 5) from running the years in pairs, on two sample paddocks, were used for the comparison, as the computing time for the *Grid* method for a continuous 10 year run would have been prohibitive. The first year of the pair is used to "prime" the model, and only the N leaching predictions from the second year are analyzed.

Results and discussion

Both methods converged when the number of months to remember was around 9–10 (Fig. 2a). Even with 10 months to remember, the *Probabilistic* method was computationally more efficient (Fig. 2b), taking less than half of the time for the 2 year runs than the *Grid* method. Furthermore, it was estimated that the *Grid* method would have taken approximately 230 days for a 10 year run (presented in section Example 2: seasonal variation in UN load) with the computer used for this study (Intel® Core™ CPU, 4300 @ 1.8 GHz, 1.79 GHz, 0.98 GB of RAM), whereas the *Probabilistic* method completed this task in less than 2 days. For a 2 year run, both methods would take more or less the same time with 14 months to remember. Ten months to remember is used throughout the rest of the study.

On runs over 8 pairs of years, using 10 months to remember on the *Probabilistic* method, both methods produced similar patterns in terms of annual N leaching [R^2 = 86 %, Mean Relative Prediction Error (MRPE) = 10 %] (Fig. 3a). The average annual N leaching was similar for the *Grid* and *Probabilistic* method (37.0 ± 14.5 vs. 38.1 ± 10.9 kg N ha^{-1} year^{-1}, respectively). In most of the years, the *Probabilistic* method predicted slightly more N leaching than the *Grid* method, except for 2008. The reasons for this inversion need to be investigated further, but it could be related to the worst

Table 3 Number of paddocks on which nitrogen leaching was monitored on Control farmlet described by Ledgard et al. (2006), and the collectors used per paddock (Data provided by AgResearch Ltd, New Zealand, Paula Phillips and Mark Shepherd)

Year	Paddocks	Ceramic cup collectors
2002	12	60
2003	10	48
2004	13	84
2005	13	84
2006	13	84
2007	14	140
2008	14	140

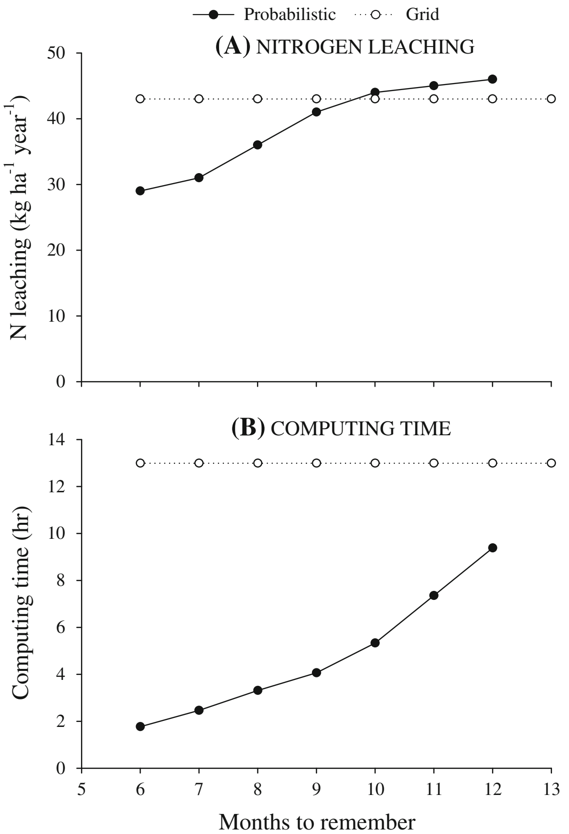

Fig. 2 Comparison between methods, *Grid* and *Probabilistic* (varying the number of months to remember), for a 2-year simulation for **a** annual N leaching and **b** computing time. Values are for one paddock out of 15 paddocks on the simulated farm and total N leaching for the second year

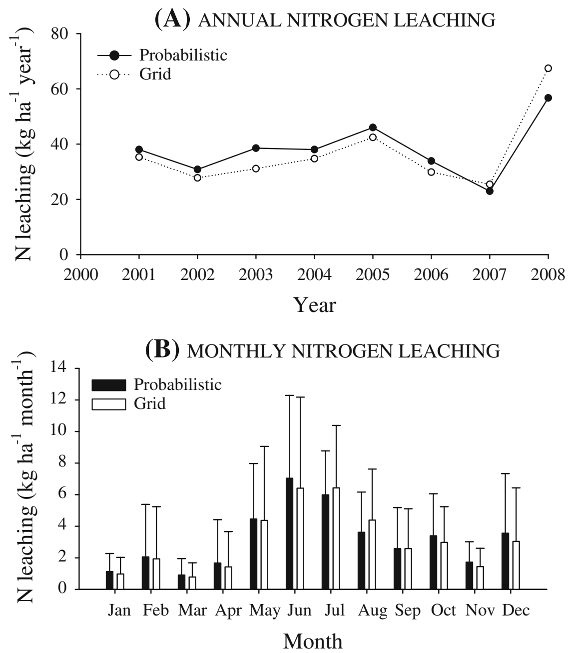

Fig. 3 Comparison between methods, *Grid* and *Probabilistic* (9 months to remember), in terms of **a** annual N leaching and **b** monthly distribution of N leaching. *Error bars* represent one standard deviation

drought on record in the Waikato region. The average monthly distribution of N leaching was also very similar between methods ($R^2 = 96$ %, MRPE = 8 %, Fig. 3b).

The comparison with observed data was satisfactory (Fig. 4), in the sense that both methods followed the correct trend reasonably well, with acceptable coefficients of determination and MRPE against observed data ($R^2 = 94$, MRPE = 10 % for *Grid* method and $R^2 = 72$ %, MRPE = 20 % for the *Probabilistic* methods). In 6 out of 7 years (86 %), predictions by both methods fell within the range of one standard deviation of the observed mean N leaching. The observed annual N leaching averaged 39.9 ± 27.9 kg N ha^{-1} $year^{-1}$, whereas the simulated results were 37.0 ± 19.6 kg N ha^{-1} $year^{-1}$ for the *Grid* method and 38.1 ± 10.9 kg N ha^{-1} $year^{-1}$ for the *Probabilistic* method. It is noteworthy that both methods showed high N leaching in 2008, which corresponded to an extremely dry summer followed by above-average winter rainfall. These results are encouraging, but this was a limited testing exercise and further research is required.

The Grid method represents a spatial simplification, whereas the *Probabilistic* constitutes a temporal simplification. There are advantages and disadvantages for each method, which are summarised in Table 4. One advantage of the *Grid* method is the ease to which functionality to include dung patches and grazing strips in the simulation could be applied. The *Probabilistic* method offers the opportunity to study longer simulation periods, with acceptable computing times.

Example 1: The importance of considering urine overlap

The addition of urine patch overlapping added considerable complexity and computational demands to the UPF model, but it is in agreement with the conclusions drawn by Shorten and Pleasants (2007) and Bates (2009). Other authors, e.g. Di and Cameron

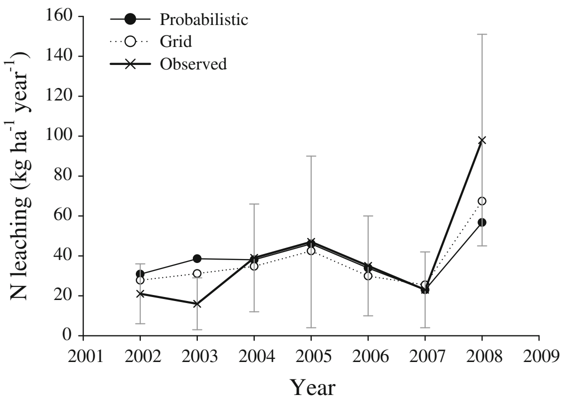

Fig. 4 Comparison between observed (control treatment resource efficient dairying trial, Ledgard et al. 2006) and simulated N leaching for the period 2002–2008, using the *Grid* method and the *Probabilistic* method (10 months to remember). *Error bars* represent the standard deviation between paddocks in the observed data

(2000), assumed overlapping of urine patches to be not important in their modeling work, although they commented that this assumption may not always hold.

Pleasants et al. (2007) concluded that, at the same overall stocking rate, grazing strategies that increase the frequency of overlapping urine patches over a short period of time (e.g. long rotation lengths in winter equates to small grazing breaks at high stocking densities) can increase the expected N leaching.

Of course a double urine patch receives twice as much urine as a single patch, and a larger proportion of the urinary N deposited can be leached. Pakro and Dillon (1995) found that the nitrogen leached from lysimeters treated with dual urinations leached 3.6 and 2.9 times more that those with single treatments (irrigated and non-irrigated paddocks, respectively). Stout (2003) measured 1.4–2.83 times more leaching, from 2 L urinations compared with 1 L urinations, depending on the time of application. In New Zealand, Di and Cameron (2007) observed quadratic (more that proportional) responses of N leaching to increases in urinary N application rates. In a modelling study, in line with the experimental evidence, Shorten and Pleasants (2007) estimated that, on average, 38 and 61 % of the nitrogen can be leached from single and double patches respectively. This implies that a double urine patch would leach about 3 times more N than a single patch, as calculated by Pleasants et al. (2007).

Figure 5 shows, as an example, the evolution of the concentration NO_3-N and NH_4-N in the soil, N uptake by the plant and NO_3-N leaching, after receiving one

Table 4 Advantages and disadvantages of the *Grid* and *Probabilistic* methods to simulate the spatial distribution of urination at paddock level

Grid method: Advantages	*Probabilistic* method: Disadvantages
Each cell runs for the full simulation period. Therefore, the soil model keeps all soil mass balances in check	*Temporal simplification*: APSIM needs to be reinitialised for each new period of time
Grazing strips and livestock-camping areas could be represented explicitly	Paddocks need to be treated as units
Soil variability can be represented	
Dung patches can be applied on the same *Grid*	Dung patches must be ignored to keep the number of cells within manageable numbers
Intuitive, easy to describe	Hard to explain, understand and debug
***Grid* method: Disadvantages**	***Probabilistic* method: Advantages**
Spatial simplification: Overlapping can only be certain proportions of a patch. In the example used here: 0, 25, 50, or 100 % (given by parameter c in Eq. 11)	All relevant urination patterns are considered
A constant area per urination needs to be assumed. This could mask the effect of animal size and dynamic changes in the volume per urination	The area per urination can be modified on each grazing event
Requires very long computing time for simulations longer than 2 years. The number of cells to simulate increases with the length of the simulation period as less pruning of identical cell is possible. Also, all cells must be run for the full simulation period	Reduces computing time, as the number of simulation-days to run in APSIM is reduced. The number of cells to run increases with the increase in the simulation time, but, by breaking time into periods, each cell runs only for a fraction (months to remember) of the full simulation period

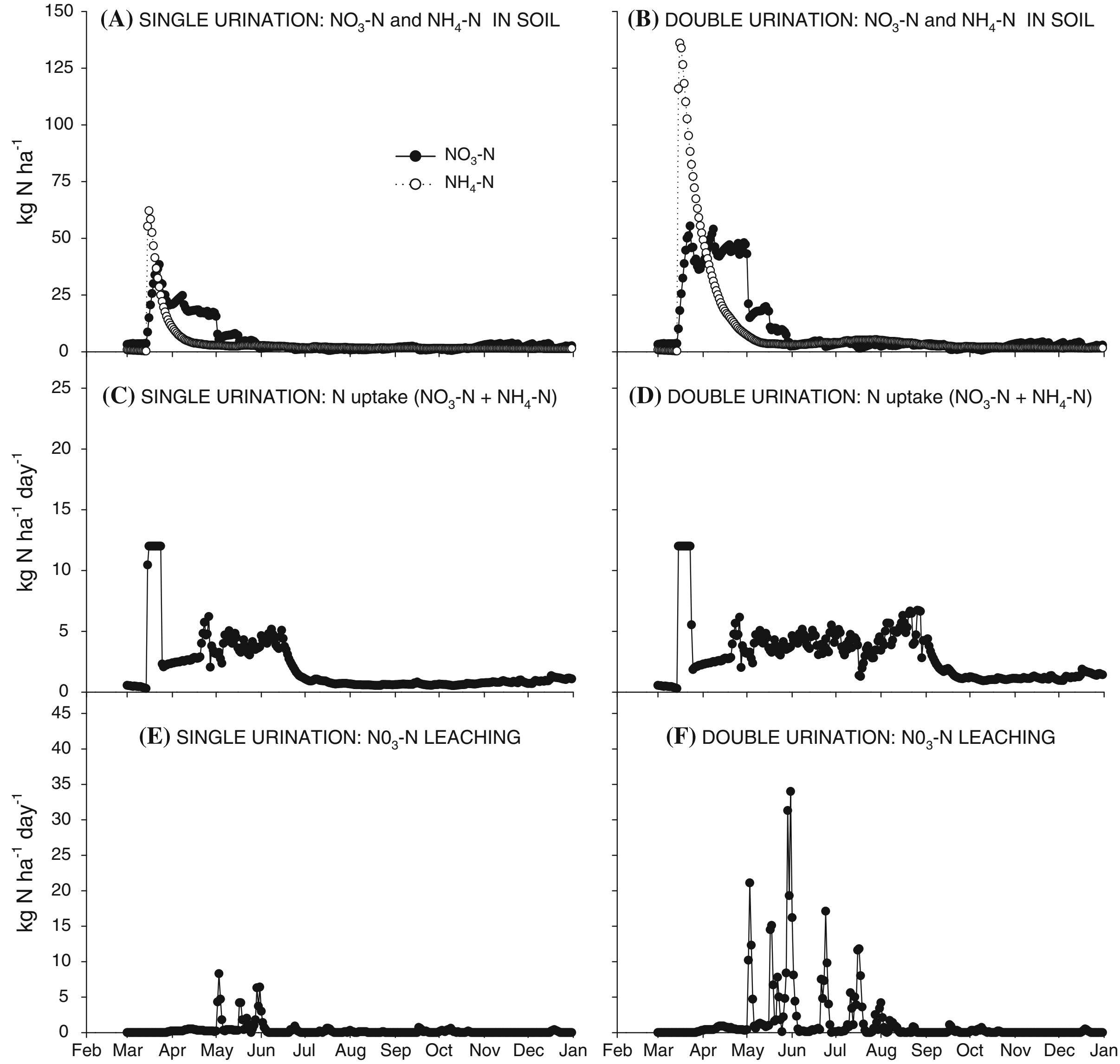

Fig. 5 Simulated NO_3-N and NH_4-N content in the top four layers of the soil (**a** and **b**), N uptake by the plant (**c** and **d**) and NO_3-N leaching below 55 cm (**e** and **f**), after a single or a double urine application on 15 March 2005

(application rate 400 kg N ha^{-1}) or two urinations on the 15 March 2005 in APSIM. In this instance, the cumulative N leaching during the winter following the application increased by about 4 times for the double compared with the single urination.

In assessing the importance of overlapping, account must be taken of the probability of a point in space being urinated on more than once, for example for 1 year the probability is low, and also the proportion of the urine volume that is deposited on an already urinated area. Using the simulated farm described in the section Model testing, and two paddocks grazed 11 and 13 times respectively during the year 2004, Fig. 6 shows the results for the proportion of the paddock area covered with urine (0–7 times), and the proportion of the total urine volume deposited with different levels of urine overlap. On average, only 8 % of the paddock area was covered with multiple urinations during 1 year. However, as much as 39 % of the total urine volume was deposited on overlapped patches, assuming uniform probability of urinations across the paddock.

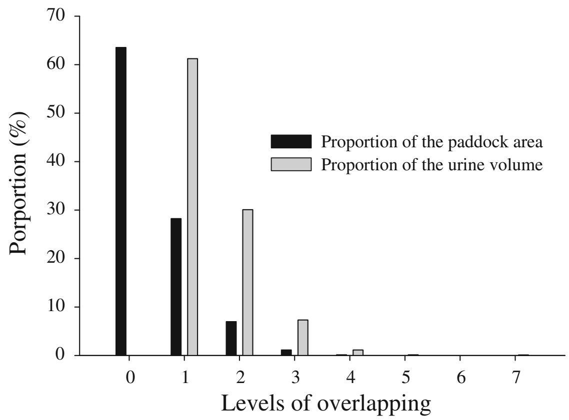

Fig. 6 Proportion of the paddock area covered by urinations (*black columns*) and proportion of the urine volume (*grey columns*) deposited with different levels of overlapping in a year. Average of two simulated paddocks

Example 2: Seasonal variation in UN load relative riskiness

The WFM parameterised to represent the farmlet described before, was run for 10 years (1999–2008). The average number of grazing events in a month (Fig. 7a) multiplied by the amount of urinary N (kg N ha^{-1}) excreted at each event (Fig. 7b) on a paddock gives the urinary N load for the paddock for each month. Figure 7c shows the average urinary N load (15 paddocks), for each month of the year, on the simulated farm. The amount of urinary N deposited at each grazing event, in turn, depends on N content in the harvested pasture (which varies with the N content and the proportion of green/dead material in the herbage) and supplementary feeds; the time the cows spend on the paddocks (as opposed to dairy and races); the amount of N diverted to milk by the cows; the stocking rate of the farm (which varies throughout the year due to culling and replacement) and the stocking density at each grazing event (which depends on grazing rotation length). For example, in August there is a momentary reduction in urinary N load onto pastures (Fig. 7c). This can be explained by the fact that August is the main calving period. Cows begin the milking season, spending less time on the paddocks (4 h for milking and 1 h on the races during milking were assumed in this study) and diverting N to milk, while intake levels (and therefore urinary N output) are still low during the post-partum period due to low herbage availability. The urinary N load increased in

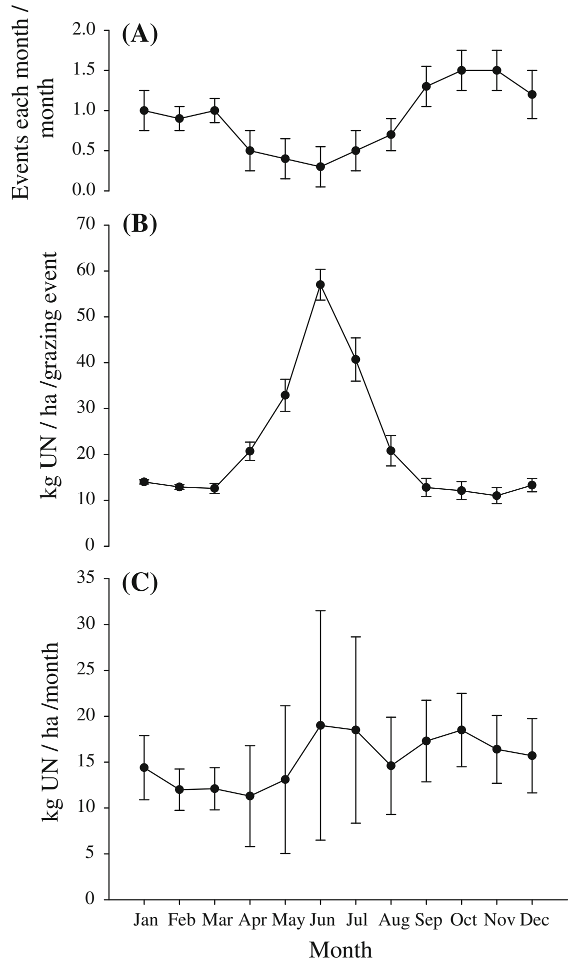

Fig. 7 Monthly simulated output (average of 15 paddocks) for **a** average number of grazing events per paddock **b** Urinary N (UN) deposited per grazing event and **c** total UN load for a simulated Waikato dairy farm on each month per year. *Error bars* represent one standard deviation (variation between years, n = 10)

May–June (Fig. 7c), and this occurs despite DM intake being lower than in spring. At that time of the year, the instant stocking density was up to 5 times higher than in spring (rotation length 100 days in June, and 20 days from September to December), the cows were non-lactating thus diverting no N to milk and they spend 100 % time on pasture and the replacement cows enter the herd at the end of May.

Restricting grazing time, by keeping cows on standoff pads to prevent excreta being deposited onto pastures during critical times, is one way of reducing N leaching (de Klein and Ledgard 2001). The UPF can be used to explore the relative riskiness of urinary N depositions at different times. As an example, the urinary N loading output history from the WFM run (10 years: 1999–2008), described in the previous paragraph, was used as the input for the UPF (using only the *Probabilistic* method) to estimate N leaching from two sample paddocks out of 15 in the simulated farm. After that, grazing events (i.e. urinary N loading) were removed from the files generated by the WFM 1 month at a time. The results are shown in Table 5. A farm system level analysis must consider the effect of stand-off pads on the fate of excreta collected, which on New Zealand dairy farms is re-distributed at times of low leaching risk. In this exploratory study of relative riskiness of month of deposition, all the urinary N removed was considered to be out of the system.

Systematically removing all urinary N in May or June had a considerable effect in reducing N leaching (by about 20 %). As expected, this mitigation effect was evident when removing urinary N in any month, although the benefit was smaller in spring, when the pastures are capable of extracting a larger proportion of excess N in urine patches. Per unit of urinary N removed, removing urinary N in the first half of the year, even well before winter (the main N leaching period, Fig. 3b), was more effective than in July or August (Table 5). For the farm simulated here, removing urinary N in August had minimal benefit in reducing N leaching, which is in agreement with the results of Shepherd et al. (2010b), who measured N leaching from urine applied on different months of the year on a Horotiu soil in the Waikato region.

De Klein and Ledgard (2001) estimated a 35–50 % reduction in N leaching by completely avoiding grazing from April to August on a typical Waikato dairy farm. The effect of removing urinary N over this period was also simulated here and showed a reduction in N leaching of 50 ± 8 %. In a contrasting environment (Southland, New Zealand), de Klein et al. (2006) also determined a high N leaching mitigation potential by restricting grazing to only 3 h per day in Autumn (March–May), measuring a reduction in N leaching of 41 % compared to the standard grazing practice.

At farm system level, the most effective time for removing excreted N (by destocking/standing cows off) as an N leaching mitigation tool will depend not only on the relative riskiness, but also on the seasonal distribution of the urinary N load in relation to the distribution of rainfall, drainage and pasture N uptake characteristics of each dairy system and region. Obviously, the fate of the excreta not deposited onto the pastures would be of key importance to determine the net effect of this practice.

The simulated results reported in this paper were obtained using the default and least complex soil water module available in APSIM, and assumes a constant initial soil penetration rate of urine in the soil profile (independent of the volume of the urination event and soil moisture). Further work should be undertaken using the more complex APSIM-SWIM model. Also, rigorous evaluation of the UPF is required, with more experimental data covering different soils, climates and farm types. With these caveats in mind, the observation that urinary N deposited in summer could be also an important contributor to N leaching, which mostly happens in winter, provides an important hypothesis for further research on the design of N leaching mitigation alternatives.

Conclusions

The UPF used outputs from a whole farm model that were post-processed in APSIM and showed strong relationships between simulated and observed N leaching for a Waikato dairy farm. Further work will extend this work to include other soil types, climate and farm management settings.

The *Grid* and *Probabilistic* methods compared here are equally useful, with advantages and disadvantages that determine which one is more appropriate for different circumstances. The *Grid* method would be more appropriate when dung patches need to be represented, for example to account for soil carbon balances. The *Probabilistic* method is faster and similarly accurate when N leaching is the focus.

The results of the first example suggest that urine patch overlapping cannot simply be ignored in modelling N leaching from pastoral dairy systems from the urine patch level, at least not in all situations. This is important when comparing dairy systems with contrasting management variables, like stocking rate and rotation length, which change the frequency of urine patch overlap.

The second example indicates that avoiding urinary N deposition on pastures in the autumn or early winter could be highly effective in mitigating N leaching during the following winter.

Table 5 Average (10-year simulations) for two sample paddocks of the simulated farm for urinary N (UN) load, N leaching on each month of the year, and N leaching mitigation effect of removing UN on each month of the year a time

	Paddock	Jan	Feb	Mar	Apr	May	Jun	Jul
UN load (kg ha^{-1} month^{-1}), % in parenthesis	1	15(9)	13(7)	13(7)	10(5)	12(7)	24(13)	19(10)
	2	17(9)	15(8)	10(5)	9(5)	15(8)	20(10)	11(6)
N leaching (kg ha^{-1}month^{-1}), % in parenthesis	1	0.9(2.4)	1.6(4.3)	0.5(1.5)	1.6(4.2)	4.9(13)	6.7(17.6)	6.1(16.1)
	2	1.0(2.8)	1.5(4.2)	0.8(2.1)	2.0(5.5)	4.4(12)	6.9(18.9)	5.7(15.7)
Average N leaching reduction when removing urinary N on each month of the year								
Reduction in annual N leaching (kg N ha^{-1} year^{-1})	1	4.8	5.1	5.6	2.4	7.3	7.5	4.7
	2	2.5	3.7	4.1	2.5	6.3	6.2	1.5
Reduction in annual N leaching (%)	1	10.6	11.8	13.2	7.0	17.0	19.9	9.7
	2	8.5	9.5	10.1	6.3	17.1	19.4	4.1
N Leaching reduction efficiency (kg N leaching saved/kg UN removed)	1	0.31	0.39	0.42	0.25	0.59	0.31	0.25
	2	0.15	0.25	0.40	0.27	0.41	0.32	0.14

	Paddock	Aug	Sep	Oct	Nov	Dec	Annual
UN load (kg ha^{-1} month^{-1}), % in parenthesis	1	12(7)	16(9)	16(9)	15(8)	14(8)	181
	2	18(10)	17(9)	18(10)	21(11)	16(8)	186
N leaching (kg ha^{-1}month^{-1}), % in parenthesis	1	3.9(10.3)	3.2(8.3)	3.7(9.9)	1.9(5.1)	2.7(7.2)	37.82
	2	3.5(9.6)	2.6(7.2)	3.5(9.6)	1.8(4.9)	2.7(7.5)	36.48
Average N leaching reduction when removing urinary N on each month of the year							
Reduction in annual N leaching (kg N ha^{-1} year^{-1})	1	1.2	3.5	4.1	4.0	4.5	
	2	2.2	2.1	2.9	2.7	3.7	
Reduction in annual N leaching (%)	1	2.2	8.6	9.8	6.8	3.2	
	2	5.7	6.5	8.5	6.6	8.7	
N Leaching reduction efficiency (kg N leaching saved/kg UN removed)	1	0.10	0.21	0.26	0.27	0.31	
	2	0.12	0.13	0.16	0.13	0.24	

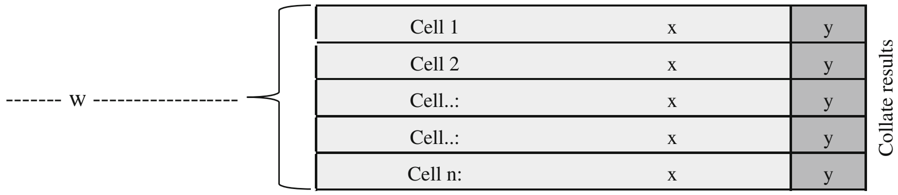

Fig. 8 Schematic representation of the sequence of APSIM runs for one period of time in the Probabilistic method. *w* represents the initialisation period of the whole paddock, x the initialisation period of each cell and y is the period when N leaching data are collected

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Appendix

In the *Probabilistic* method, each time "period" involves three simulation stages (Fig. 8). APSIM is first run once for each paddock for W months to generate initialisation values for the state variables NO3, NH4, water in the soil and soil organic matter, which are stored. APSIM is then re-initialised using these values and run for each cell for X months to remember + Y months of actual output collection. Finally, the results for each cell are collated and the N leaching is calculated as:

$$\mathrm{Nleaching}_{y} = \sum_{i=0}^{n} \mathrm{Nleaching}_{y,i} \times \frac{\mathrm{area}_{y,i}}{\mathrm{paddockArea}}$$

where $\mathrm{Nleaching}_y$: total N leaching during period y from the whole paddock (kg N ha^{-1}); $\mathrm{Nleaching}_{y,i}$: N leaching during period y from the ith cell (kg N ha^{-1}); $\mathrm{area}_{y,i}$: size of the ith cell (ha).

The length of the three periods (W, X and Y) can be modified by the user.

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